Math 152 Problem Set 7

Due 11/20/2026, at 10:00 PM

Problem 1 (Optional)**.** Consider a cone with base radius r and height h. Imagine we drill a hole of radius $r/2$ through the cone along the vertical axis of symmetry.

(a) Sketch the resulting solid, including a side-on view.

(b) Find the area of a horizontal cross section at height y above the base.

(c) Compute the volume of the solid.

(d) Describe the solid as a solid of revolution.

Problem 2. The Voyager spacecraft are powered by radioisotope thermoelectric generators that generate electricity from the heat produced as Plutonium 238 decays. This isotope has a half-life of 87.7 years, and each generator starts with a 4.5kg sample.

(a) Find the decay constant of Plutonium 238, and find a formula for the mass $m(t)$ remaining (in kg) after t years. You may use a calculator for computations here.

(b) Write down the differential equation satisfied by $m(t)$.

(c) Given that the Voyager spacecraft have been operating for about 50 years, estimate how much of the Plutonium 238 remains in each one. Again, feel free to use a calculator.

Problem 3. Evaluate the following integral:

$$\int_{-1}^{1} \frac{2-x}{4+(x-1)^2}\mathrm{d}x$$

Problem 4. Suppose that $\theta \in [0, \pi/2)$. Use the unit circle to determine how each of the following angles is related to θ:

(a) $\sin^{-1}(\sin(\pi+\theta))$

(b) $\tan^{-1}(\tan(\pi-\theta))$

(c) $\sin^{-1}(\cos(\theta))$

Problem 5. In this problem we will work out a geometric interpretation of the formula

$$\int f(x)\mathrm{d}x = xf(x) - \int xf'(x)\mathrm{d}x$$

in a special case. Suppose that f is an increasing invertible function with $f(0) = 0$. Using the above formula to evaluate the integral of f from 0 to a, we have

$$\int_0^a f(x)\mathrm{d}x = a \cdot f(a) - \int_0^a xf'(x)\mathrm{d}x$$

(a) Sketch the graph of an example of such an f from 0 to a (i.e any increasing function starting from the point $(0, 0)$.) Interpret the first two terms in the above formula as areas of regions in your sketch.

(b) Determine that the third term should also represents the area of a region on your sketch. Use the change of variable formula[1] to rewrite it as an integral in y, and explain why the resulting integral represents the area that it does.

[1] You will need to use the inverse function f^{-1}

Problem 6. Let n be a positive integer. Find a general formula for the following indefinite integral:

$$\int \exp(x)\sin(nx)\mathrm{d}x.$$

Problem 7. Following the example of the natural logarithm[2], use integration by parts as in Problem 5 to find the following antiderivatives:

(a) $\int \sin^{-1}(x)\mathrm{d}x$

(b) $\int \tan^{-1}(x)\mathrm{d}x$

[2]This will be done in class on Monday